Graphs 9 — Gradient of a function · Differentiation · Stationary points · Motion in a straight line

Calculator allowed. Marks in brackets. ★ = harder question. If $y = kx^n$ then $\dfrac{dy}{dx} = nkx^{n-1}$.

1 Find the gradient of the tangent to each curve at $x = 1$. [7]

a $y = 2x^3 + 5x - 1$ **b** $y = (x - 1)(4x + 3)$ **c** $y = 3 - \dfrac{4}{x^2}$

2 Find the stationary points of the curve $y = x^3 - 6x^2 + 9x + 1$ and say whether each is a maximum or a minimum. [6]

3 The fuel consumption y km per litre of a car travelling at x km/h is modelled by $y = 60 - \dfrac{x}{4} - \dfrac{1600}{x}$ for $40 \le x \le 120$. **a** Find an expression for $\dfrac{dy}{dx}$. **b** Find the most economical speed and the fuel consumption at that speed. [6]

4 The displacement s metres of a particle from a fixed point O after t seconds is $s = 6\sqrt{t} + 2t + 5$ for $0 \le t \le 4$. Find, when $t = 1$, the particle's **a** velocity in m/s **b** acceleration in m/s². [6]

Total: 25 marks

Check: multiply by the power, then reduce the power by 1; constants vanish · expand brackets or rewrite $\frac{a}{x^n}$ as ax^{-n} and $\sqrt{x}$ as $x^{\frac{1}{2}}$ before differentiating · stationary point: $\frac{dy}{dx} = 0$; classify by gradient either side · velocity $= \frac{ds}{dt}$, acceleration $= \frac{dv}{dt}$.